GECAD – RESEARCH GROUP ON INTELLIGENT ENGINEERING AND COMPUTING
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2025 Competition on Electric Energy Consumption Forecast Adopting Multi-criteria Performance Metrics

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1. Description

This dataset is the official release of data for the 2025 Competition on Electric Energy Consumption Forecast Adopting Multi-criteria Performance Metrics. The competition is open and welcomes everyone who wishes to participate and to anyone who can benefit from these data.

2. Competition Outline

Forecasting of electric energy consumption can be a very difficult task when handling building-level data. However, an accurate forecast is needed to boost the potential of energy management systems (EMS). The need to forecast energy consumption grows as our reliance on renewable energy sources, such as solar and wind power, grows. This means that to meet consumer demand with renewable energy generation, energy management systems must operate based on accurate energy forecasting models for both short and long-term periods.

Energy consumption forecasting techniques that can manage a variety of scenarios, including varying prediction timeframes, accessible data, data frequency, and even data quality, have been the subject of intense research. There is no one-size-fits-all approach, where certain situations call for different approaches. The goal of this competition is to compile and evaluate the most recent advances in energy consumption forecasting techniques.

3. Releases Details

- **Release v1.0:** One full year of operational telemetry from a smart building with readings taken every 5 minutes.

- **Release v2.0:** 40 days of data from a smart building with readings taken every 5 minutes.
- **Release v3.x:** A single day of data from a smart building with readings taken every hour (First competition phase: from 06/01/2025 to 10/01/2025).
- **Release v4.x:** A single day of data from a smart building with readings taken every hour (Second competition phase: from 14/07/2025 to 18/07/2025).

4. Dataset Description & Measured Variables

All data releases are composed of the following synchronized telemetry channels:

- **Time:** Timestamp recorded in hours and minutes.
- **Power:** Total building electric power consumption in Watts (W).
- **Voltage:** Line voltage in Volts (V).
- **Current:** Electric current draw in Amperes (A).
- **Generation Power:** Local rooftop Photovoltaic (PV) solar power generation in Watts (W).
- **Temperature:** Ambient outdoor temperature in degrees Celsius (°C).

5. Multi-criteria Performance Metrics

Participating forecasting models are quantitatively evaluated across multiple dimensions:
1. Predictive Accuracy: Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE);
2. Computational Performance: Model execution time for training and inference;
3. Computational Energy Efficiency: Energy required by the computational platform during forecasting execution.